

Track 2 Best Model Collage Report

Track 2 Visual Best-Model Comparison

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report compares representative **Track 2** TE-curve predictions for the current best reference, Track 1, Wave 1 directional, and Wave 1 global models. Each model is shown as one four-image collage so local oscillation tracking can be inspected directly.

Scope

- each collage contains four deterministic held-out test curves;
- forward models are shown on forward curves only;
- backward models are shown on backward curves only;
- global Wave 1 models are shown on two forward and two backward curves;
- Measured TE** uses the same line width as predictions and a dark-gray color for balanced visual comparison.

Metrics Summary

Forward Reference Best Models

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
paper_original_best_Fw	rcim_original	Fw	0.002769	0.002951	6.250
paper_retuned_best_Fw	rcim_retuned	Fw	0.001839	0.002041	4.109
track1_best_Fw	rcim_track1	Fw	0.003014	0.003204	6.819

Forward Wave 1 Family Best Models

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
feedforward_fw	wave1_current_registry	Fw	0.003404	0.003855	7.551
harmonic_regression_fw	wave1_current_registry	Fw	0.003230	0.003494	7.185
periodic_mlp_fw	wave1_current_registry	Fw	0.003391	0.003814	7.524
residual_harmonic_mlp_fw	wave1_current_registry	Fw	0.003273	0.003563	7.266
tree_fw	wave1_current_registry	Fw	0.003053	0.003395	6.731

Backward Reference Best Models

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
paper_retuned_best_Bw	rcim_retuned	Bw	0.003675	0.004284	7.572
track1_best_Bw	rcim_track1	Bw	0.005027	0.005212	11.860

Backward Wave 1 Family Best Models

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
feedforward_bw	wave1_current_registry	Bw	0.003586	0.004023	7.832
harmonic_regression_bw	wave1_current_registry	Bw	0.003678	0.004012	8.058
periodic_mlp_bw	wave1_current_registry	Bw	0.003574	0.004006	7.807
residual_harmonic_mlp_bw	wave1_current_registry	Bw	0.003536	0.003874	7.728
tree_bw	wave1_current_registry	Bw	0.003258	0.003651	7.051

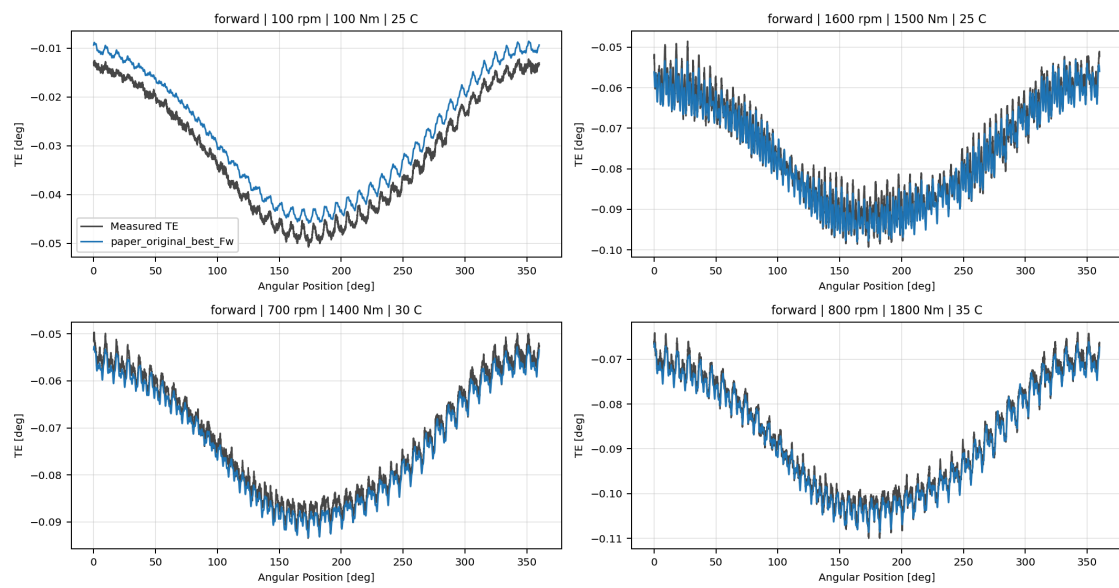
Global Wave 1 Family Best Models

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
feedforward_global	wave1_current_registry	global	0.003465	0.003897	7.636
harmonic_regression_global	wave1_current_registry	global	0.018129	0.018330	41.458
periodic_mlp_global	wave1_current_registry	global	0.003447	0.003872	7.582
residual_harmonic_mlp_global	wave1_current_registry	global	0.003407	0.003822	7.486
tree_global	wave1_current_registry	global	0.003144	0.003533	6.854

Collage Gallery - Forward Reference Best Models

[paper_original_best_Fw](#)

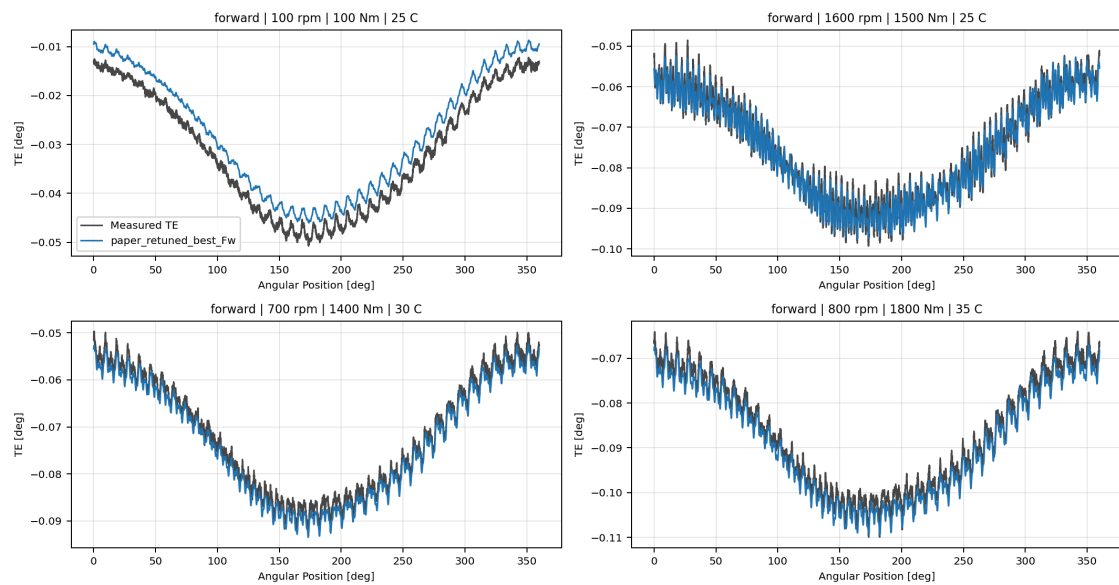
paper_original_best_Fw



paper_original_best_Fw Track 2 collage

[paper_retuned_best_Fw](#)

paper_retuned_best_Fw

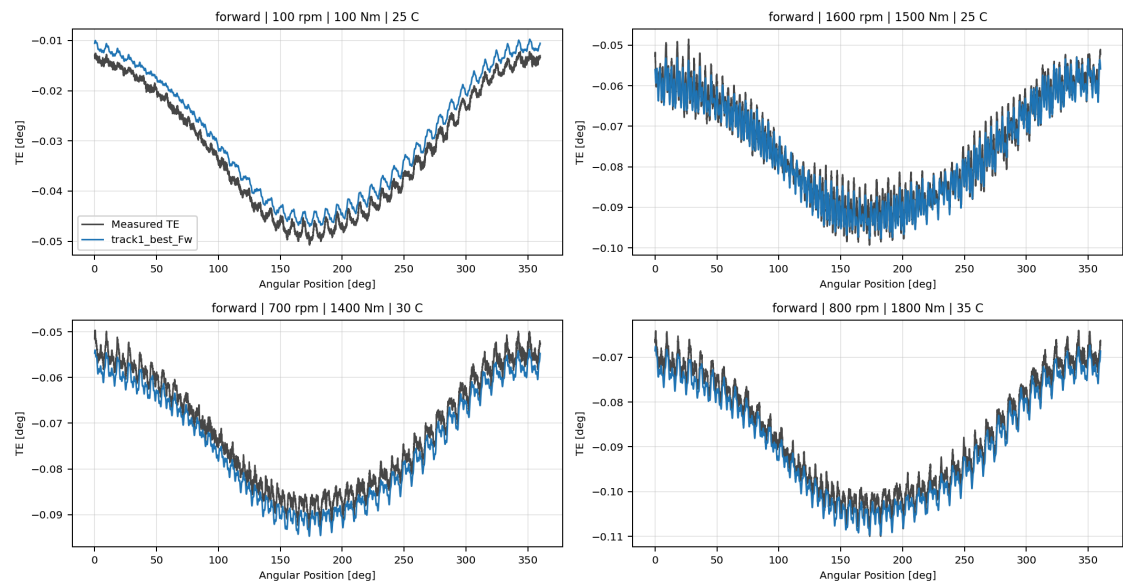


paper_retuned_best_Fw Track 2 collage

Collage Gallery - Forward Reference Best Models Continued

track1_best_Fw

track1_best_Fw

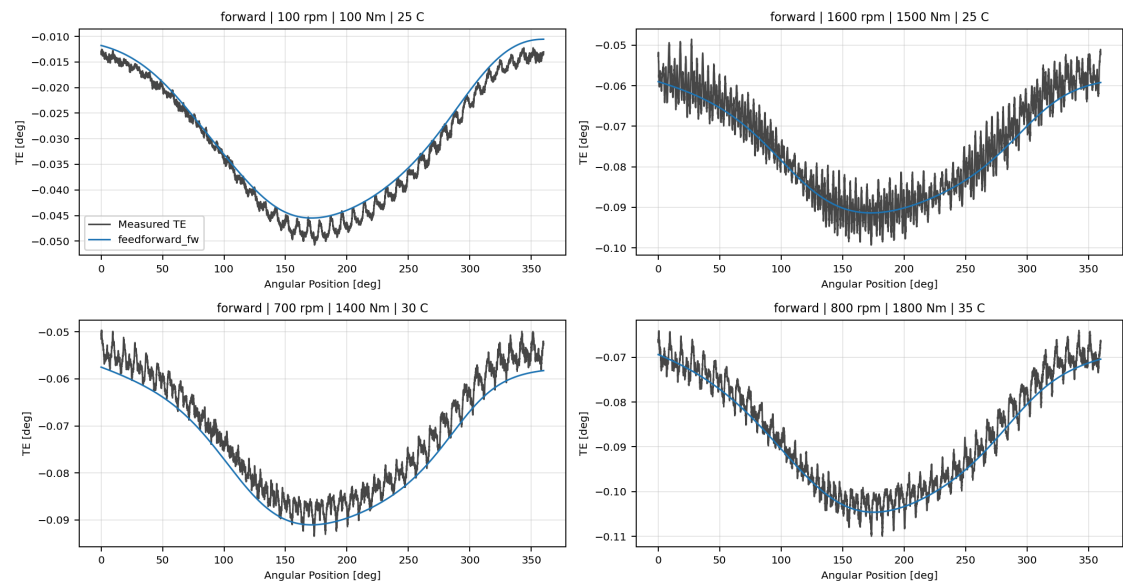


track1_best_Fw Track 2 collage

Collage Gallery - Forward Wave 1 Family Best Models

feedforward_fw

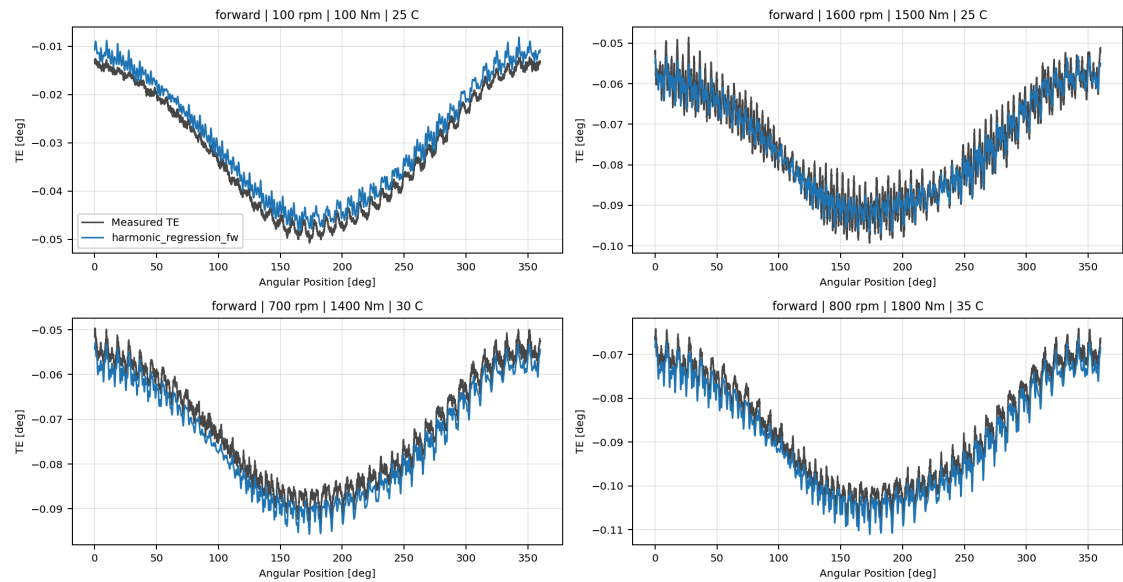
feedforward_fw



feedforward_fw Track 2 collage

harmonic_regression_fw

harmonic_regression_fw

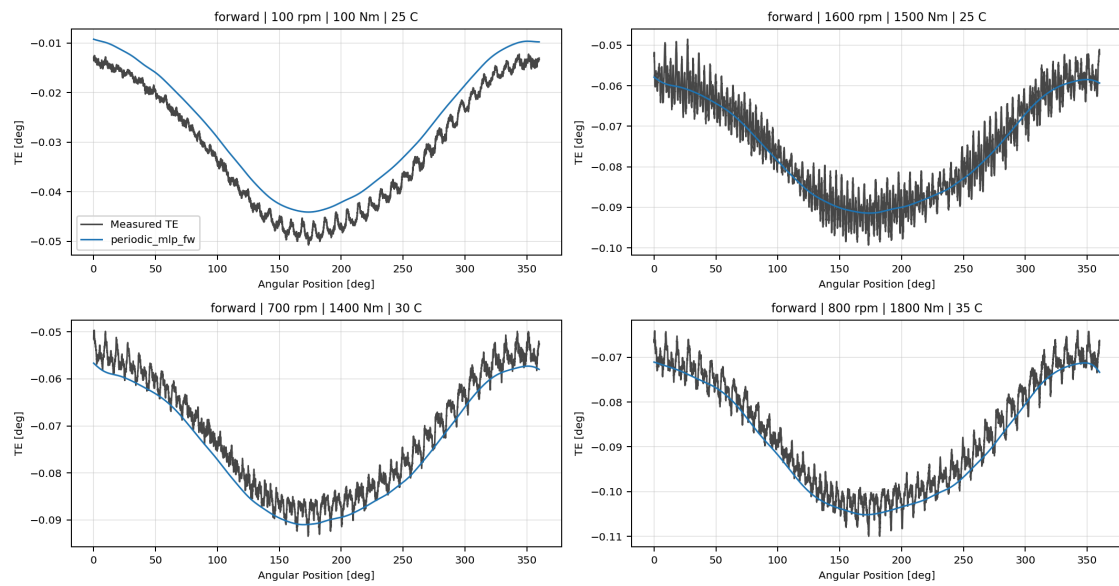


harmonic_regression_fw Track 2 collage

Collage Gallery - Forward Wave 1 Family Best Models Continued

periodic_mlp_fw

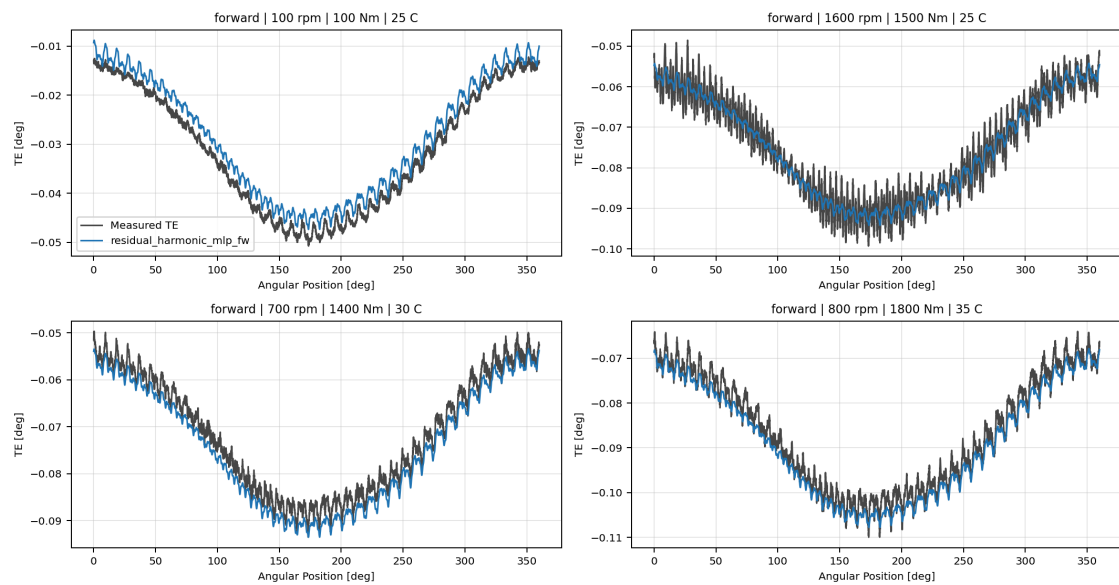
periodic_mlp_fw



periodic_mlp_fw Track 2 collage

residual_harmonic_mlp_fw

residual_harmonic_mlp_fw

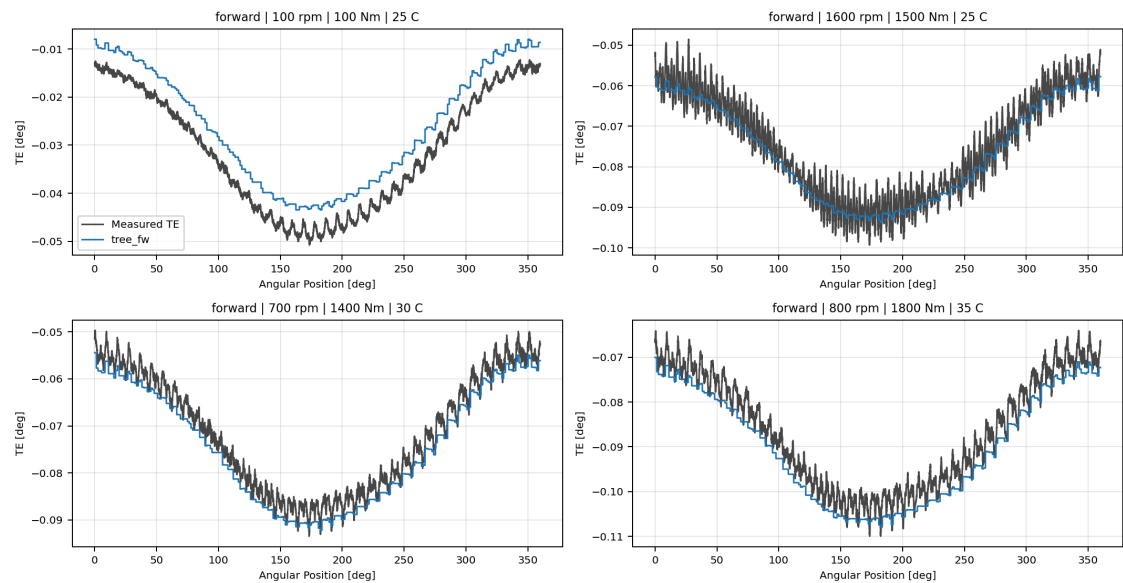


residual_harmonic_mlp_fw Track 2 collage

Collage Gallery - Forward Wave 1 Family Best Models Continued 2

tree_fw

tree_fw

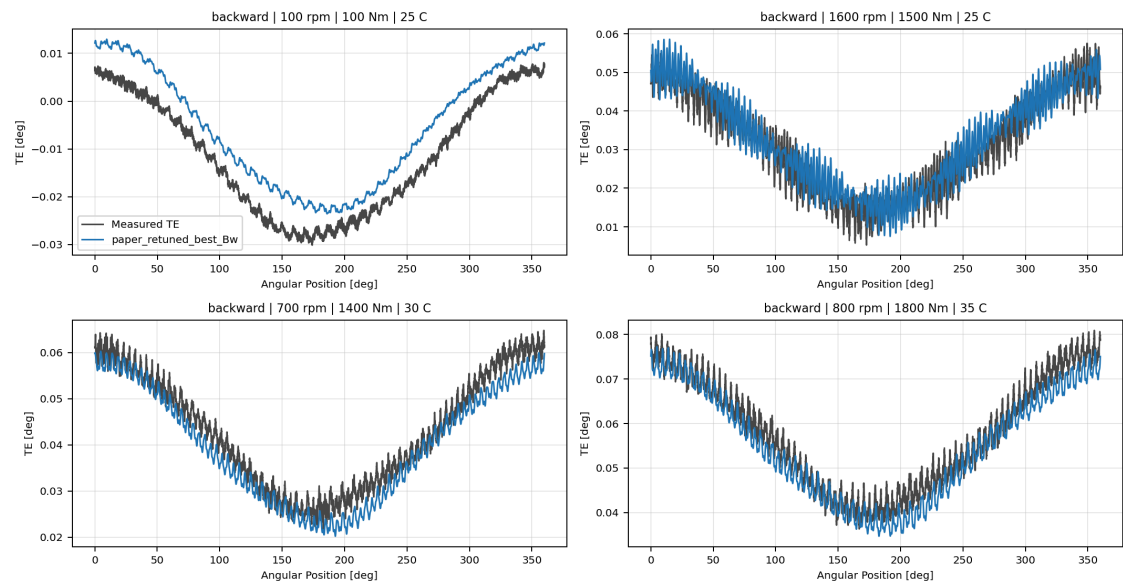


tree_fw Track 2 collage

Collage Gallery - Backward Reference Best Models

paper_retuned_best_Bw

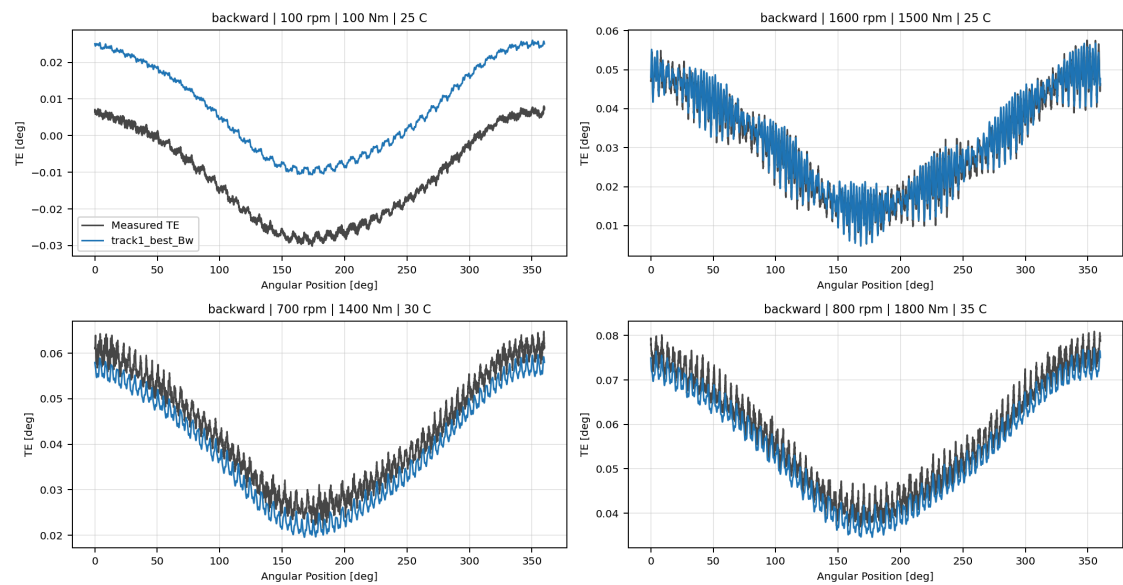
paper_retuned_best_Bw



paper_retuned_best_Bw Track 2 collage

track1_best_Bw

track1_best_Bw

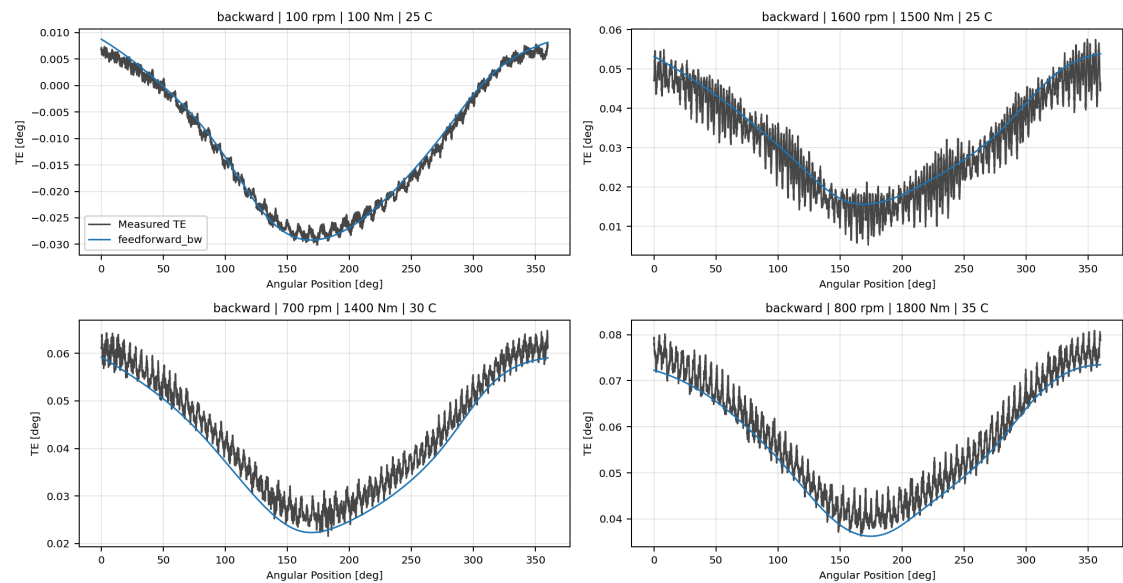


track1_best_Bw Track 2 collage

Collage Gallery - Backward Wave 1 Family Best Models

feedforward_bw

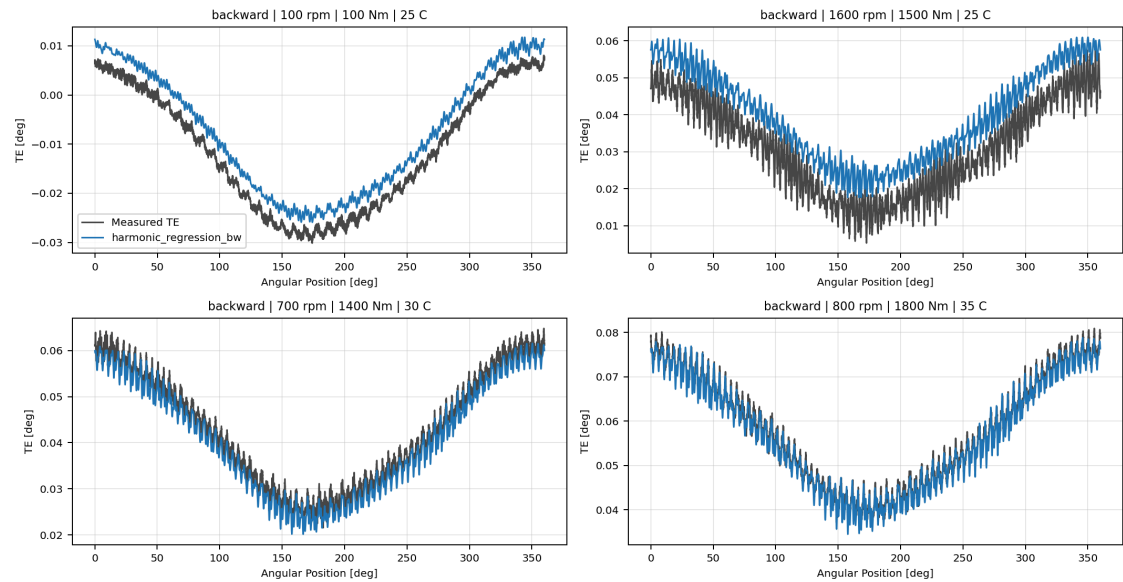
feedforward_bw



feedforward_bw Track 2 collage

harmonic_regression_bw

harmonic_regression_bw

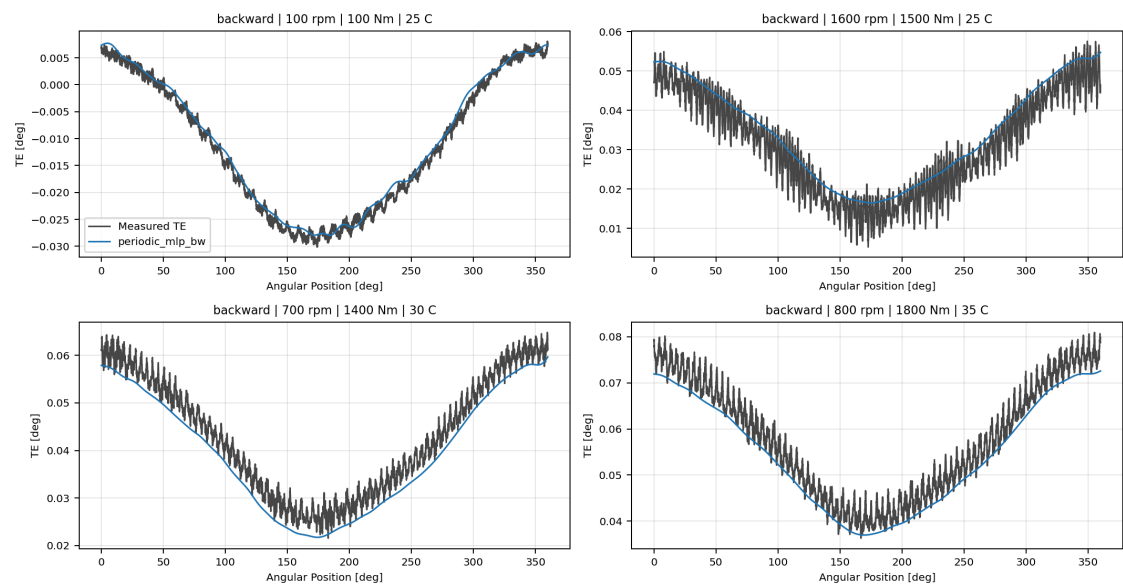


harmonic_regression_bw Track 2 collage

Collage Gallery - Backward Wave 1 Family Best Models Continued

periodic_mlp_bw

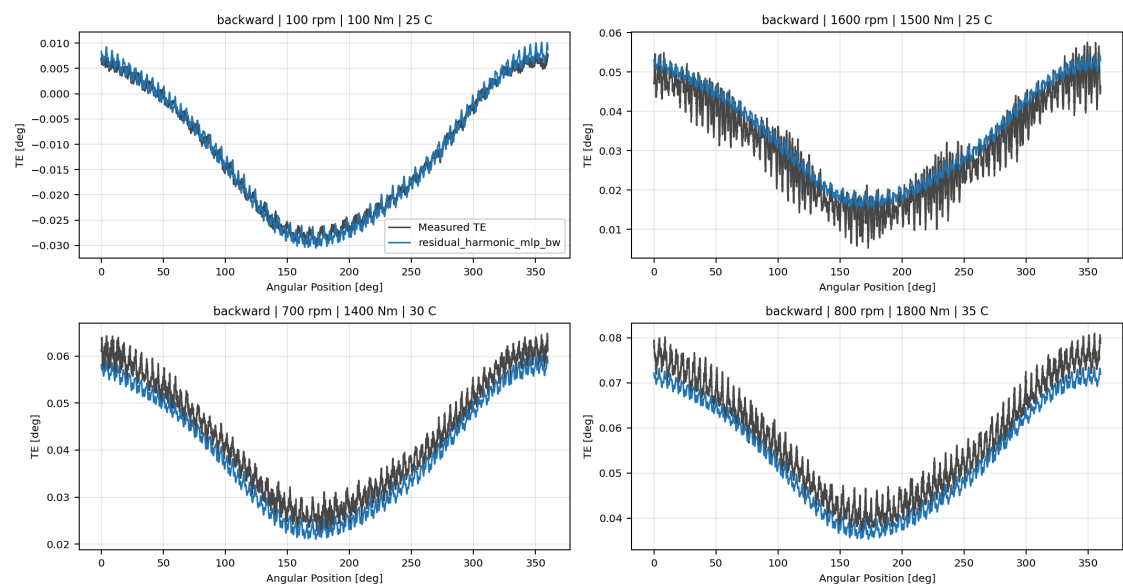
periodic_mlp_bw



periodic_mlp_bw Track 2 collage

residual_harmonic_mlp_bw

residual_harmonic_mlp_bw

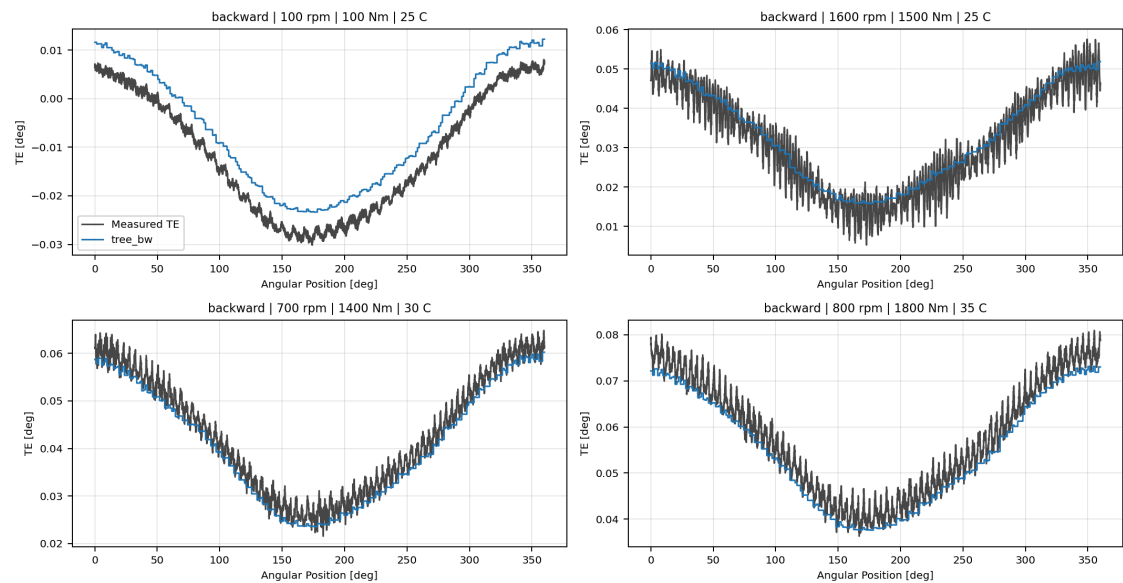


residual_harmonic_mlp_bw Track 2 collage

Collage Gallery - Backward Wave 1 Family Best Models Continued 2

tree_bw

tree_bw

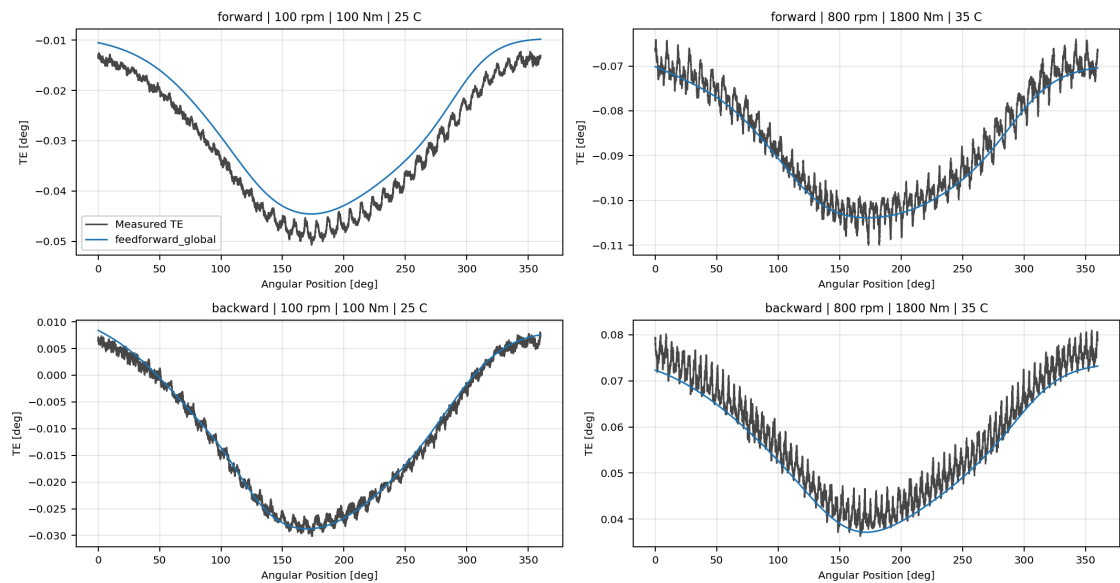


tree_bw Track 2 collage

Collage Gallery - Global Wave 1 Family Best Models

feedforward_global

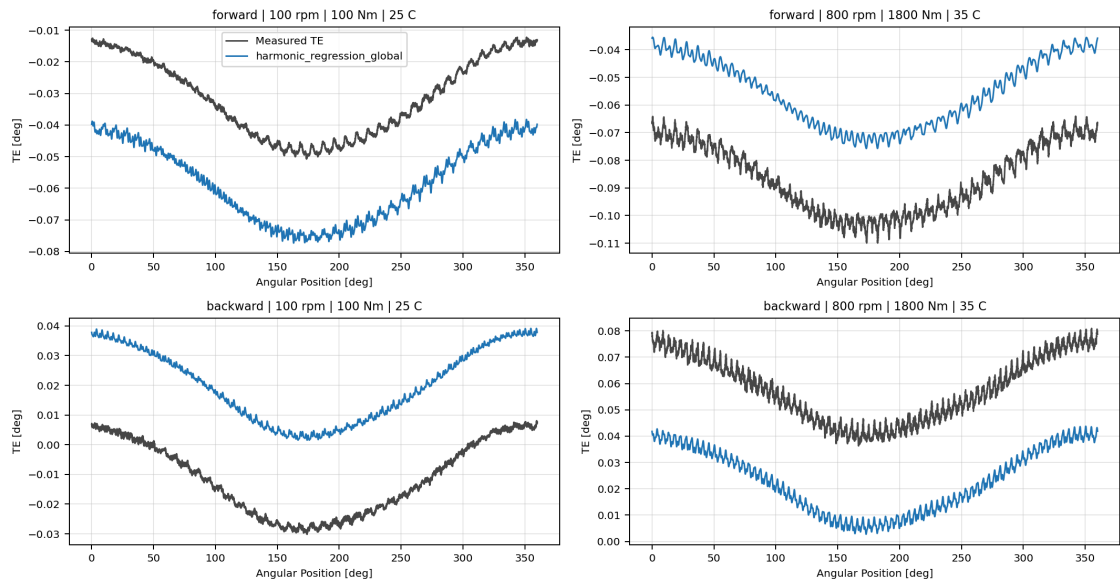
feedforward_global



feedforward_global Track 2 collage

harmonic_regression_global

harmonic_regression_global

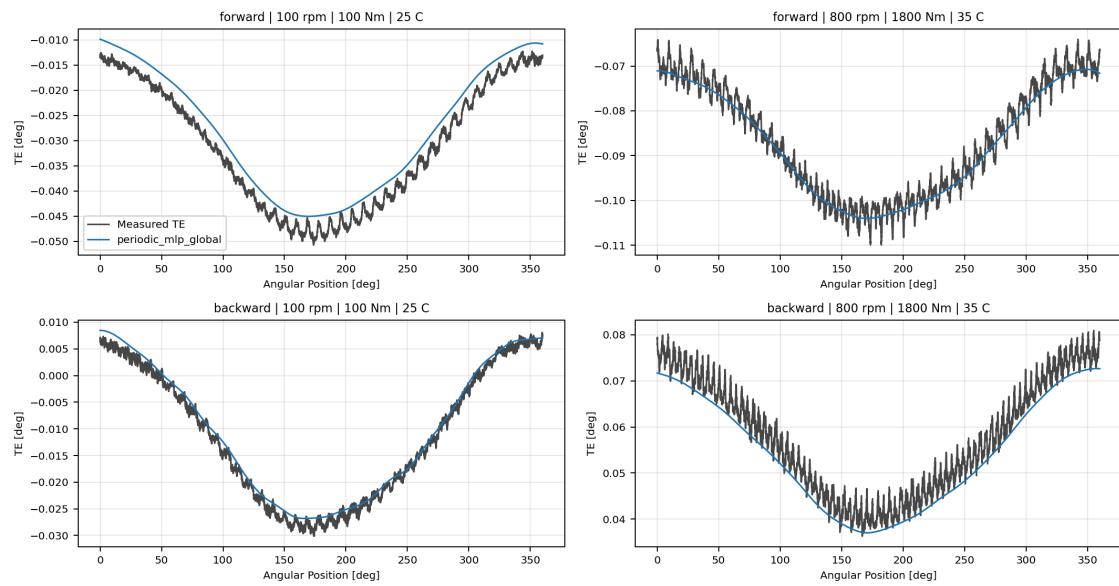


harmonic_regression_global Track 2 collage

Collage Gallery - Global Wave 1 Family Best Models Continued

periodic_mlp_global

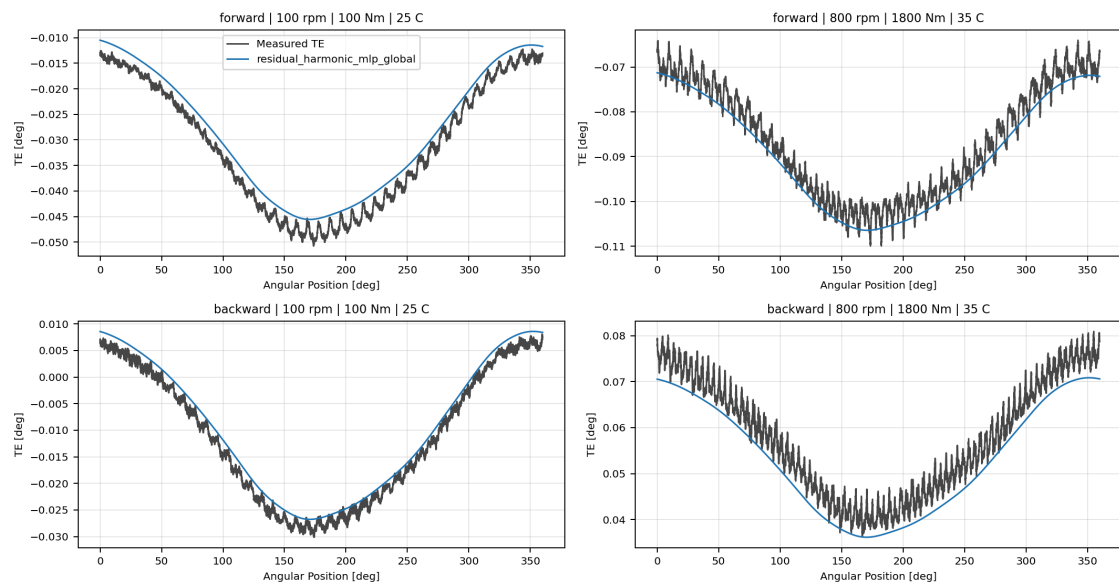
periodic_mlp_global



periodic_mlp_global Track 2 collage

residual_harmonic_mlp_global

residual_harmonic_mlp_global

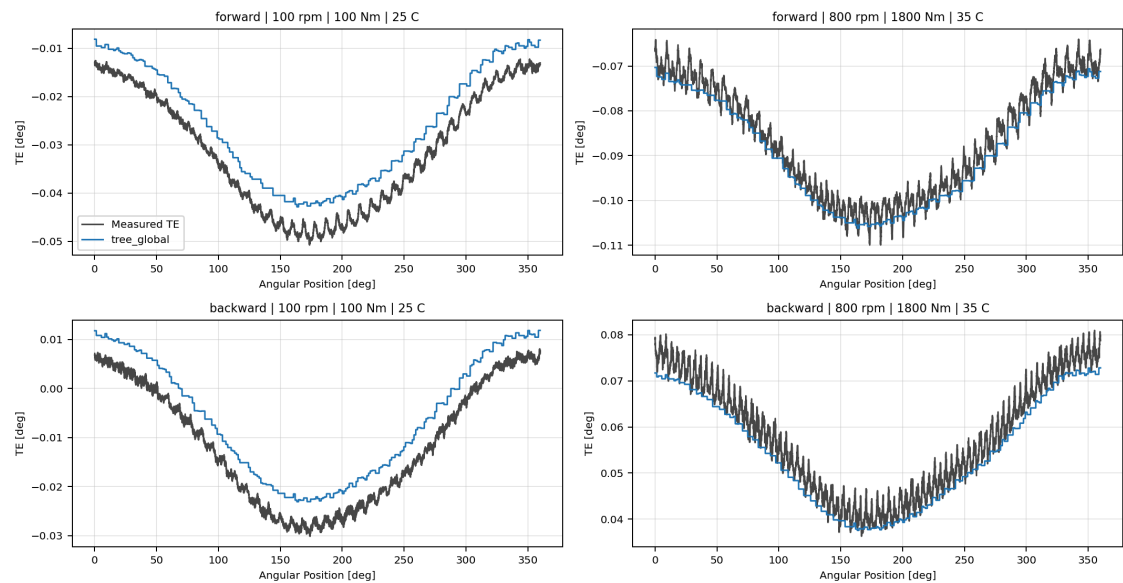


residual_harmonic_mlp_global Track 2 collage

Collage Gallery - Global Wave 1 Family Best Models Continued 2

tree_global

tree_global



tree_global Track 2 collage

Output Artifacts

- output directory: `output\validation_checks\track2_best_model_collage_report\2026-05-20-19-56-41__track2_best_model_collage_report` ;
- summary YAML: `output\validation_checks\track2_best_model_collage_report\2026-05-20-19-56-41__track2_best_model_collage_report\track2_best_model_collage_summary.yaml` ;
- metrics CSV: `output\validation_checks\track2_best_model_collage_report\2026-05-20-19-56-41__track2_best_model_collage_report\track2_best_model_collage_metrics.csv` ;
- report Markdown: `doc\reports\analysis\track2\best_model_collage_report\[2026-05-20]\track2_best_model_collage_report.md` .